Data Science
##
Manifesto
    wrangle cyclopean tentacled chaos
    tame C'Thu'Lu to domestic service

Data easily becomes overwhelming.
Alien character compositions establish a barrier insurmountable
    -- to all but the faithful intelligent.

Dirty data is often a clean structure malformed for the current readership.

The first mission is to transform data into a human readable form.

Plan of action:
    What is the end purpose, expected-output, dimension?
    Who created the data, human or automation?
    Where was the data encoded?
    When was the time of creation; can time enable segregation?
    Why was the data created and how was it created? Does such reason match your purpose?

WORKFLOW:
1. Purpose
2. Data source & data elements
3. Sanatize data and discover data structure
4. Explore & Analyze data
5. Algorithm to structure data
6. Verify results
7. Reports

###
Remember

When the task is unsurmountable. Breathe. The first step is to be passive.
Watch the data. Print the data, element by element at a readable pace.
Let it become familiarized in the background.
Slices of data from random, start, mid, end - let a relationship develop.

First, one must be well-acquainted, and that takes time.
Patiently view the data while the system is designed.

Afterwards, the structure will be revealed.
Global dimensions which are able to contain any particular object.
Hierarchy to be flattened by classification.
Quarantine unstructured anonmaly to be analyzed last.
Work with objects that can be easily contained, and grow from that stable center.
Do not try to contain all under universal dominion.

Identify patterns and relationships within the data.

Look for trends, correlations, and anomalies.
Apply operations to extract insights by trial & error.

Finally, apply appropriate techniques to extract insights and achieve the end purpose.

Never forget that automation is the endgame.
Hands-on data grooming is essential until the system is mastered.
###
Purpose

Establish the smallest working set, of actions requiring interaction, by means of automata

1) Read
2) Reduce
3) Transform
4) Standardize
5) Verify
6) Automata

###
Core Ideas

FILTERS:
    Automata discrimination structures - to transform, join, exclude with the object to Sanitize & Standardize.
    Gatekeepers of what may be considered as input to the system.
    Logs & discards unacceptable inputs.

POOLS:
    Collections of standardized objects of a system.
    The trusted-working source for the system.

MAPS:
    Explicit relationships designed for Operations, amongst elements & structures.
    Maps are built as the byways of pools to structure.

STRUCTURES:
    Preliminary, transactional, and final forms requisite for the system.
    Final form used by the system to compose results.
###
Element Reduction

Goal is to reduce the element into its irreducible unique form by operations which:
    { prune, compress, encode }

First analyze the whole, to obtain a minimum-structure which all fulfill.
Next consider the nature within the whole dimensions, seek ways to operate towards reduction.
Never colapse elements into collision of identity, each element must remain unique.

If several components can be collapsed into a single classification
    -- Then supplant the components with a single encoding.
The encoding should be an essential distinction.

Nurture the virtue of Simplification, Clarity, & Efficency.
Redundancy is a burden.
##
Data Migration

Data often exists in an origin encoding
    which first must be extracted & preprocessed
    only after can the filters sanitize & standardize

Proprietary exports are hostile territories.
The major effort is to transform the data into a workable pool without loss of information.
##
LOGIC:
set
sequence
group

OPERATION:
if then
for
while
switch
matrix
remaindering (chisel towards)

REGEX:
pattern theory
unique tag identifiers (pattern match target)
match
search
replace
concatenate
anchors
net capture
formulae
chaining

STRUCTURE:
cell
array
hash
array of arrays
hash of arrays of arrays
object

multi-set ledger
nested arrays, arrays with related sub-arrays

PURPOSE:
comparison
reduction
filtering
sanitizing
reading encodings
sorting
aggregation

OUTPUT:
text
table
graph
pdf
email

BENCH:
shell
python
pandas
spreadsheet
modules

DEEP:
linear systems
tuple theory
matrix algebra

OTHER:
exception handling
type checking
file encoding

SPACES:
raw binary space - unix terminal util
forge of logic engines - c
logic of situation - perl python
spreadsheet information logic - pandas
matrix numerical information - numpy
multiset ledger reconcile information
graphic projection - data plotting
report processing - pdf
stream of data - data is not retained, only shapes a image state or alerts upon event
structural/related tabular data, unisolated structures - sql, trees

Programming Theory:
Blackbox Abstraction
Longhand

Programming falls into two schools of reasoning.
Bluecollar usage of general purpose tools.
Whitecollar construction of tools.

Accumulation of expertise of the one does not accummulate the other.
These are two independent paths which establish contrary frames of existence.
Proficient manipulation VS hacking creation.

Blackbox Abstractions allow war tested immediate modules to process in an ideal for the GENERAL situation. The tool of use lives in an entire ecosystem of the module.
This nested entity is not easily modified, less so understood.

Longhand keeps the hands of the creator in the clay. the logic is bare, there exists a flow and readership. Portability is supreme because the logic is explicitly simple.
The code is long. what may be a simple call in a Blackbox, may be the most complex machinery.

Blackbox Abstraction builds the bluecollar worker mindset: exist within the acceptable realm, and be entirely powerless when the situation does not fit.

Longhand builds the whitecollar mindset: exist entirely unlimited except by time.
Powerful in all situations.